Advanced Data Analytics - 216H03C701

Course Outcomes

- Design and implement frequent itemset mining and clustering algorithms to discover patterns and groupings in large transactional and streaming datasets using memory-efficient techniques.
- Apply stream mining techniques to analyze and summarize continuous data using efficient algorithms for sampling, filtering, and counting.
- Apply tree mining frameworks to extract structured patterns from complex hierarchical data.
- Apply text mining and recommendation techniques using vector space models, classification algorithms, and dimensionality reduction methods.
- Apply sequence and time series mining techniques to discover patterns and trends in multivariate temporal datasets.

ADA Teaching Plan

Module 1 (3 Hours) Frequent Itemset Mining & Clustering

- Introduction to Data Mining
- KDD Process
- Market Basket Analysis
- Apriori Algorithm
- Frequent Item Mining
- Stream Mining
- CURE Clustering
- Parallel Clustering

Module 2 (6 Hours) Mining Data Streams

- Stream Data Model
- Reservoir Sampling
- Filtering
- Counting Distinct Elements
- Sliding Window
- Decaying Window
- Moment Estimation

ADA Teaching Plan

Module 3 (10 Hours) Mining Complex Structures

- Tree Mining
- Ordered Trees
- Unordered Trees
- TMG Framework
- Closed Frequent Trees
- Maximal Trees
- Applications

Module 4 (8 Hours) Text Mining & Recommendation Systems

- Vector Space Model
- TF-IDF
- Naïve Bayes
- Bernoulli Model
- Feature Selection
- Text Classification
- Collaborative Filtering
- Content-Based Recommendation
- Dimensionality Reduction

ADA Teaching Plan

Module 5 (8 Hours) Sequence Mining & Time Series Mining

- Sequential Pattern Discovery
- Bioinformatics Applications
- Multivariate Time Series
- Dynamic Time Warping (DTW)
- Generalized Sequential Pattern Algorithm

Modes of Content Delivery

- Resource Sharing
  - Google Classroom/Moodle LMS
- Some time Online Teaching – Learning
  - Gmeet/MS Teams platform / Zoom
- IA submission

Modes of Content Devlivery

- Visual Aids
- Assignments
- Seminar
- Video Lectures
- Quiz
- Guest Lecture

Reference Books

Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
Hadzic F., Tan H. & Dillon T. S	Mining data with Complex Structures	Springer	2011
Yates R. B. and Neto B. R	Modern Information Retrieval	Pearson Education	2005
Han J. & Kamber M	Data Mining: Concepts and Techniques	Morgan Kaufmann Publishers	Third edition, 2011
Christopher D.M., Prabhakar R. & Hinrich S	Introduction to Information Retrieval"	Cambridge UP	Online edition, 2009
Tan P. N., Steinbach M & Kumar V	Introduction to Data Mining	Pearson Education	2006

Evaluation Scheme

- Number of credits – 04 (TH – 03 , PR – 01)
- Mid Semester Exam (MSE) – 30 marks
- Internal Assessment – 20 marks (Two IAs)
- End Semester Exam – 50 marks

Mid Semester Examination

- Frequent Itemset and Clustering (Module 1)
- Mining Data Streams (Module 2)
- Mining complex structures (Module 3)

Internal Assessment

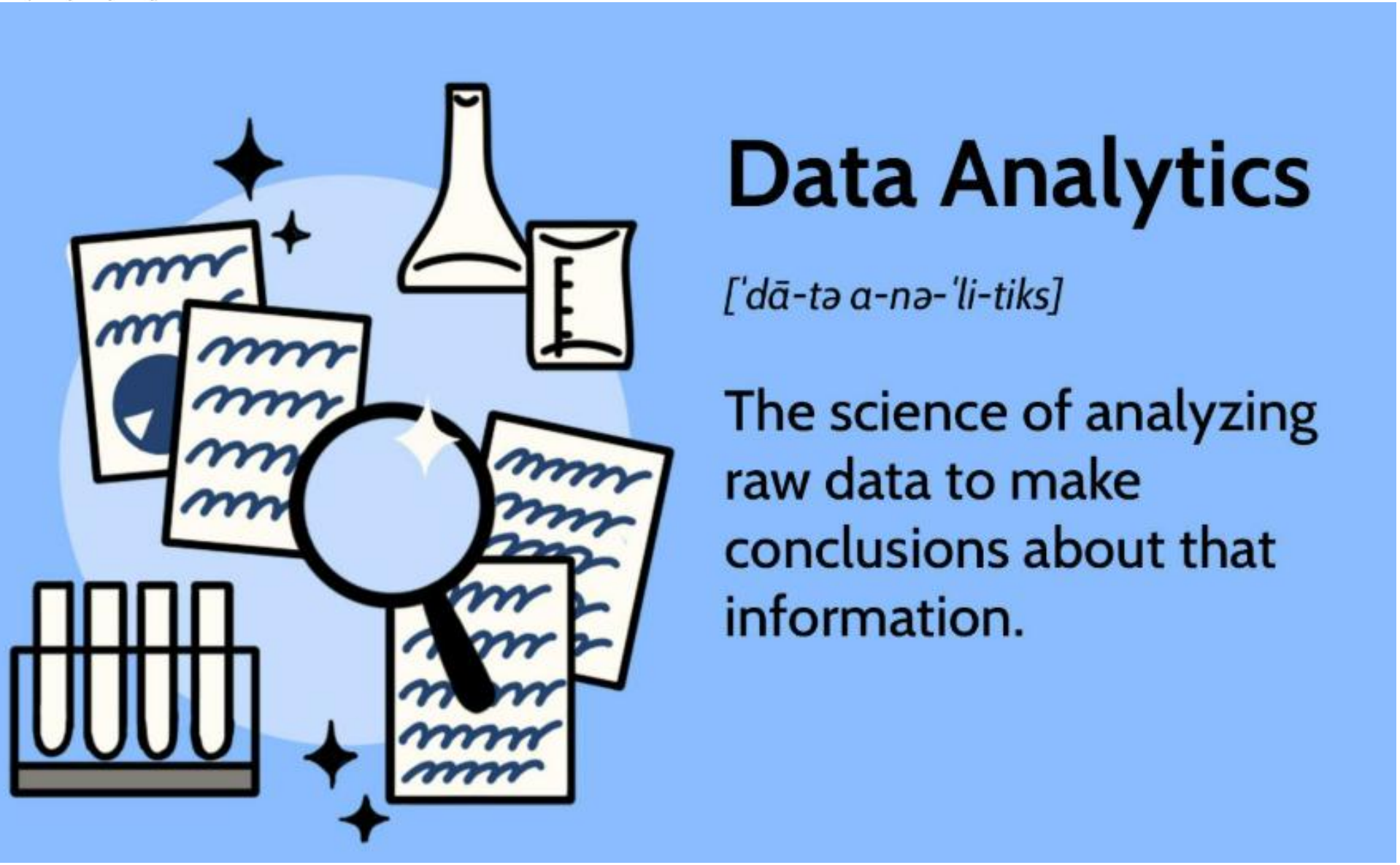
- IA 1 - Quiz 20 Marks
  - Only one quiz on LMS after MSE.
- IA 2 – Industry based Case Study 20 Marks
  - Group of 4 students
  - Poster Preparation and recording

Real World Applications

Domain	Practical Syllabus Application	Industry Value
Healthcare	Sequence Mining & Tree Frameworks	Finding mutation paths in genetic sequences and parsing structural patient EHR charts.
Finance & Security	Streaming Decaying Windows & Filtering	Tracking suspicious credit sweeps, flag pattern deviations, and isolating distributed denial attacks.
Smart Cities / Climate	Multivariate Time Series (MVTS) & DTW	Spatio-temporal parsing of heatwaves, rainfall, and grid distribution variables (IMD Case Study).
E-Commerce & Retail	Market Basket Models & Recommenders	Dynamic catalog generation, layout optimization, and personalized real-time user discounts.
Social Media	Text Analytics & Clustering Engines	Processing text feeds to extract real-time trends and sentiment classification models.

Career Paths

- **Data Engineer / MLOps Engineer:**
  - Building the infrastructure for stream processing, decaying windows, and automated feature pipelines.
- **Machine Learning / AI Engineer:**
  - Developing recommendation systems, embedding models, and custom time-series predictors.
- **Data Scientist / BI Analyst:**
  - Uncovering deep user archetypes, itemsets, and advanced customer behavior trends.


Data Analytics
['dā-tə ɑ-nə-'li-tiks]
The science of analyzing raw data to make conclusions about that information.

Data analytics vs. advanced data analytics

	Basic data analytics		Advanced data analytics	
	Descriptive analytics	**Diagnostic analytics**	**Predictive analytics**	**Prescriptive analytics**
Scope	Summarizes past data. "What happened?"	Identifies reasons behind past outcomes. "Why did it happen?"	Forecasts future trends and outcomes. "What's likely to happen?"	Recommends actions based on predictions. "What should we do?"
Focus	Reporting and tracking metrics (e.g., sales figures, website traffic).	Root cause analysis, correlations, anomalies.	Anticipating customer behavior, market trends, or risks.	Optimizing decisions, simulating scenarios, prescribing strategies.
Techniques	Aggregation, averages, percentages, trend analysis.	Statistical analysis, correlation, variance analysis.	Machine learning algorithms (e.g., regression, decision trees), time series analysis.	Optimization models, scenario simulations, reinforcement learning.